Al Liu

Rust Systems Engineer & Open-Source Maintainer

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PROFILE

Rust systems engineer and open-source maintainer specializing in performance-sensitive infrastructure, asynchronous networking, distributed protocols, storage engines, and concurrency. Experienced in building reliable production systems and developing reusable Rust infrastructure libraries.

PROFESSIONAL EXPERIENCE

Senior Rust Systems Engineer

Jan 2026 - Present

Confidential Financial Technology Company

Full-time / Remote

- Develop and maintain performance-sensitive Rust infrastructure for real-time financial systems.
- Work on asynchronous networking, concurrent data pipelines, and distributed data processing.
- Improve production reliability, latency, and resource efficiency.

Rust Engineer

May 2024 - Jul 2024

Freenet

Part-time / Remote

- Contributed to decentralized networking infrastructure in freenet-core and freenet-stdlib.
- Worked on node initialization, configuration, hostname resolution, and transport-key handling.
- Improved event-log reliability and workspace maintainability.

Rust Software Engineer - Nomos

Nov 2022 - Mar 2024

Status

Full-time / Remote

- Developed simulation tooling for the Carnot distributed consensus algorithm.
- Contributed to infrastructure supporting the Nomos Layer 1 blockchain.
- Worked on privacy-preserving networking, including mixnet implementation.

Rust and Go Engineer

Jan 2022 - Sep 2023

Polar Sync

Full-time contract / Remote

- Developed Rust procedural macros and code-generation infrastructure.
- Worked on GraphQL, SQL, and The Graph subgraph functionality.
- Designed reusable abstractions for database and query integration.

Open-source engineering

Selected Rust libraries for distributed systems, storage, networking, and parsing.

tokora

Deterministic parser combinators with on-demand lexing, LALR-style dispatch, explicit backtracking, and configurable diagnostics.

github.com/al8n/tokora / crates.io/tokora

smear

Zero-copy GraphQL lexing, parsing, and validation, with reusable combinators and an optional Sans-I/O execution core behind the proto feature.

github.com/al8n/smear

memberlist

SWIM membership and failure detection with Lifeguard extensions, a Sans-I/O protocol core, and pluggable runtime and transport layers.

github.com/al8n/memberlist / crates.io/memberlist

serf

Gossip-based discovery and coordination with tunable convergence and support for multiple async runtimes.

github.com/al8n/serf / crates.io/serf

fs4

Cross-platform file locking, allocation, and filesystem utilities with async support. A maintained fork of fs2-rs using rustix.

github.com/al8n/fs4 / crates.io/fs4

skipdb

An embedded in-memory MVCC database with zero-copy storage and optimistic or serializable snapshot isolation.

github.com/al8n/skipdb / crates.io/skipdb

Explore the full project collection

TECHNICAL EXPERTISE

Languages Rust / Go / SQL
Systems & networking Linux / TCP / UDP / QUIC / WebSocket / Async I/O / Protobuf
Distributed systems SWIM / Gossip / Service discovery / Consensus / Fault tolerance
Storage & performance MVCC / Concurrent data structures / Skiplist / Caching / Zero-copy / Memory-mapped I/O
Additional technologies WebAssembly / WASI / GraphQL

EDUCATION

University of Auckland
Bachelor of Science with Honours in Computer Science
2019 - 2022 / Graduated November 2022